

Reaction Time and the Distracted Driver

U01 / S01.4

NAME _____

DATE _____

PERIOD _____

GROUP MEMBERS _____

GOALS

- Measure the distance a falling meter stick travels before you can catch it, to the nearest 0.1 cm.
- Calculate a reaction time from that distance using the kinematic equation for free fall.
- Compare your reaction time across three attention conditions and find which costs you most.
- Determine how far a car travels while its driver is still deciding to brake.
- Record every drop as you take it — including the ones you throw out, and why.

PURPOSE

You are going to measure your own reaction time with nothing but a meter stick and the fact that gravity is dependable — then do it twice more, once with your eyes closed and once with your eyes open and your brain busy. The last number you calculate today is a stopping distance, and it is the one that matters.

BACKGROUND

Free fall does not just mean "falling." An object is in **free fall** when gravity is the only force acting on it. A dropped meter stick qualifies — air resistance on something that dense, that slow, over that short a drop is far too small to detect with a ruler. A sheet of paper does not. The misstep nearly everyone makes is assuming the heavier object falls faster. It does not. **Mass does not appear anywhere in the equations you are about to use**, which says it plainly.

Gravity adds 9.80 m/s of speed every second. Near Earth's surface the acceleration due to gravity has a magnitude of $g = 9.80 \text{ m/s}^2$. That is an acceleration — a rate of change of velocity — not a speed and not a force. After 0.20 s of falling the stick is moving about 2.0 m/s; after 0.40 s, about 3.9 m/s. It never stops speeding up the whole way down.

Sign convention for this lab: down is negative. Declare it once and use it in every line of every calculation. Then the acceleration is $a = -9.80 \text{ m/s}^2$, and a stick that falls 18.4 cm has a displacement of $\Delta y = -0.184 \text{ m}$. Both are negative, they divide to a positive number, and the time comes out positive — which it must, because clocks do not run backwards. If you end up taking the square root of a negative number, you mixed your signs. You did not break physics.

DISPLACEMENT WITH CONSTANT ACCELERATION

$$\Delta y = v_0 t + \frac{1}{2} a t^2 \quad \text{meters, meters per second, seconds}$$

The stick starts at rest, and that is what makes this simple. At the instant Student A lets go the stick is not moving, so $v_0 = 0$, the entire first term is zero and disappears, and rearranging what is left for time gives you the only equation you need today.

FREE-FALL TIME FROM DROP DISTANCE (valid only when $v_0 = 0$)

$$t = \sqrt{(2\Delta y \div a)} \quad \text{seconds}$$

The meter stick is a clock. It is not testing your grip or your arm speed — it turns a time you have no instrument to measure into a distance you can. Everything between the stick leaving Student A's fingers and your fingers closing — light reaching your eye, your brain recognizing it, the signal going down your arm — happens in free fall. **The further it falls, the longer you took.**

PERCENT ERROR

$$\text{percent error} = (| \text{experimental} - \text{accepted} | \div \text{accepted}) \times 100$$

Accepted value for Situation A: 0.20 s. Published simple visual reaction times for adults run roughly 0.18–0.25 s. Know what that is before you use it: a **typical value for people**, not a constant of nature like g . A large percent error here means you are unusual, not that you measured badly — and your error analysis has to say which one you think it is.

SAMPLE CALCULATION — WORK YOURS EXACTLY LIKE THIS

A group averages their four drops and gets **18.4 cm**. Find the reaction time, then the percent error.

- Convert to meters, put the sign on it, and list what you know.** The equation runs in meters, not centimeters, and the stick fell *down*.
 $18.4 \text{ cm} \times (1 \text{ m} \div 100 \text{ cm}) = 0.184 \text{ m} \rightarrow \Delta y = -0.184 \text{ m}$
 $\Delta y = -0.184 \text{ m} \cdot v_0 = 0 \text{ m/s} \cdot a = -9.80 \text{ m/s}^2 \cdot t = ?$
- Start from the full equation and cancel the term you know is zero.**
 $\Delta y = v_0 t + \frac{1}{2} a t^2 \rightarrow \Delta y = \frac{1}{2} a t^2$
- Solve for t before you put numbers in.**
 $t^2 = 2 \Delta y \div a \rightarrow t = \sqrt{(2 \Delta y \div a)}$
- Substitute, and watch the units cancel.** Meters over meters per second squared leaves seconds squared; its square root is seconds.
 $t = \sqrt{(2(-0.184 \text{ m}) \div (-9.80 \text{ m/s}^2))} = \sqrt{(0.03755 \text{ s}^2)}$
- Answer, with a unit and the right significant figures.** 18.4 cm has three, so t has three.
 $t = 0.194 \text{ s}$
- Percent error against the accepted 0.20 s.**
 $(|0.194 \text{ s} - 0.20 \text{ s}| \div 0.20 \text{ s}) \times 100 = 3\%$
 The subtraction leaves 0.006 s — **one** significant figure — so the percent error gets one. Writing 3.0% claims a precision you do not have.

Look at the square root before you go further. A stick falling 18.4 cm takes 0.194 s; one falling four times as far takes only twice as long. Reaction time does not scale with distance the way your eye wants it to, because the stick speeds up the whole way down.

PRE-LAB — DUE AT THE START OF THE PERIOD

Answer all of these in your lab notebook, numbered 1 through 8, before you walk in. You will not be allowed to start without the pre-lab finished — most of these are exactly what people get wrong when they wing it.

- In your own words, what does **free fall** mean? Name the force we are ignoring and the specific reason ignoring it is reasonable here.
- A 50 kg and a 90 kg student each drop an identical meter stick from the same height. Do the sticks land at different times? Justify it with acceleration — not weight.
- Explain why the distance the stick falls measures your **reaction time** and not your grip strength.
- Write down the sign convention you will use today and the acceleration with its sign attached. Every calculation this period has to agree with what you just wrote.
- Read Part A.** What quantity do you measure, with what tool, to what precision?
- Read Part A, step 6.** List the three things that void a drop. For each, say whether keeping it would make your average reaction time look *faster* or *slower* than it is.
- Read Part B.** Student C has a job in Part B that nobody has in Part A. What is it, and what exactly is Student C supposed to be watching?
- Set up your three data tables in your notebook. Copy them exactly as they are printed on the DATA page, in ink, with room to write in every cell. **Units go in the column headings — do not plan to add them later.**

MATERIALS

Per group of three: meter stick (1) · desk or table with a flat front edge · calculator (1) · lab notebook and pen, one per person

PROCEDURE

Roles. A is the **dropper**, B is the **catcher** — the one being measured — and C is the **caller and recorder**. Everyone is the catcher for four drops in every situation, so you rotate three times per part.

Part A — Situation A: Ordinary Reaction Time

- B sits with a forearm flat on the desk and the hand hanging past the front edge, thumb and index finger open about 3 cm apart.
 - The fingers must not touch the stick. If they do, B feels the release instead of seeing it and the number is worthless.
- A holds the meter stick vertically and hanging, so the **0 cm end is at the bottom** and level with the top of B's thumb.
 - If your stick reads 100 at the bottom, turn it over — otherwise every number you record is wrong by the length of the stick.
- A says "ready," then releases at a random moment within five seconds. **No countdown, no "three, two, one," no twitch of the arm first.**
- B squeezes the instant the stick moves — close the fingers where they already are, do not chase the stick down.
- C reads the mark level with the **top of B's thumb**, to the nearest **0.1 cm**. That number is the distance the stick fell.
- Repeat until B has **four good drops**. A drop is **void** if B chased the stick down, if B guessed the release and caught early, or if the stick went past B's hand entirely.
 - Void it, write down *why*, and take a replacement drop. Do not quietly keep a bad number and do not quietly delete one.
- Rotate until all three of you have four good drops.

Part B — Situation B: The Passenger's Warning EYES CLOSED

Same setup as Part A, with two changes: B's eyes are shut, and C now triggers the catch.

- B closes both eyes and keeps them closed until the catch, hand in the Part A position.
- C stands where the stick is clearly visible and **watches the stick itself**, not A's hand or face.
- A releases with no warning. C shouts **"GO!"** the instant the stick starts to move.
- B squeezes on the word — not on a guess, and not on the sound of A's hand opening.
- After the catch, C reads the mark at the top of B's thumb, to 0.1 cm.
 - If the stick falls all the way past B's hand, **that is real data**. Record it as "missed — greater than 100.0 cm." It means the warning arrived later than a full meter of falling.
- Four good drops each, then rotate.

Part C — Situation C: Eyes Open, Mind Busy KEEP COUNTING

Setup is identical to Part A, and B's eyes stay open and on the stick the entire time — the only thing that changes is what B's brain is doing. Four good drops each, then rotate, exactly as before.

- C gives B a starting number — begin at **100** — and B counts backwards out loud **by sevens**: 100, 93, 86, 79, and so on.
- B keeps counting out loud *and* watching the stick — both at once. That is the point.
- A releases at a random moment, after B has said at least three numbers.
- B catches, and **keeps counting through the catch**. If B stops counting in order to catch, the trial is void. Take it again.
- C records to 0.1 cm, using a **different starting number** each round (100, 96, 87, 94...) so nobody memorizes the sequence.

DATA

You copied these three tables into your notebook as your pre-lab. Fill the notebook copy in **as you take each measurement** — never at the end of the period from memory. All distances to 0.1 cm. All times to three significant figures.

TABLE 1 · SITUATION A — **ORDINARY REACTION TIME** · EYES OPEN, NOTHING ELSE GOING ON

CATCHER	DROP 1 (cm)	DROP 2 (cm)	DROP 3 (cm)	DROP 4 (cm)	AVERAGE (cm)	AVERAGE (m)	REACTION TIME t (s)
Student 1							
Student 2							
Student 3							

TABLE 2 · SITUATION B — **THE PASSENGER'S WARNING** · EYES CLOSED, CATCH ON "GO!"

CATCHER	DROP 1 (cm)	DROP 2 (cm)	DROP 3 (cm)	DROP 4 (cm)	AVERAGE (cm)	AVERAGE (m)	REACTION TIME t (s)
Student 1							
Student 2							
Student 3							

TABLE 3 · SITUATION C — **EYES OPEN, MIND BUSY** · COUNTING BACKWARDS BY SEVENS

CATCHER	DROP 1 (cm)	DROP 2 (cm)	DROP 3 (cm)	DROP 4 (cm)	AVERAGE (cm)	AVERAGE (m)	REACTION TIME t (s)
Student 1							
Student 2							
Student 3							

If you write a number wrong, draw **one line** through it and write the correct value beside it. Never scribble it out and never erase — I need to see what you originally recorded. A voided drop gets the same treatment plus one word saying why.

WORKING YOUR DATA — DO THIS BEFORE YOU LEAVE

1. Average the four distances for each person in each situation, in centimeters, then convert to meters.
2. Calculate the reaction time for each average using $t = \sqrt{(2\Delta y \div a)}$. Three significant figures, work shown in your notebook.
3. For **Situation A only**, calculate your percent error against the accepted 0.20 s.
4. **Write your spread** for Situation A — highest drop minus lowest drop, in centimeters. That spread is your uncertainty, and you need it for Question 3.
5. Return every meter stick to the rack unbent and leave the desk clear.

POST-LAB ANALYSIS

Answer these in your lab notebook, numbered 1 through 4. Show every calculation with units. An answer with no work and no units earns partial credit at best.

- Rank and compare.** Using your own three reaction times, rank the situations from fastest to slowest. Then calculate how much longer your Situation C time was than your Situation A time — first in seconds, then as a percent increase, using $(C - A) \div A \times 100$. State in one sentence what you were doing differently in Situation C, given that your eyes were open and on the stick both times. **PARTS A & C**
- The chain.** Situation B was almost certainly your slowest, and the obvious explanation — "sound is slower than sight" — is **wrong**. Published measurements have people reacting to a sound about 0.02–0.04 s *faster* than to a light. Explain what else had to happen in Situation B that did not in Situation A, name whose reaction time is buried inside your Situation B number besides your own, and say why B is slow anyway. Then connect it: a passenger yelling "stop!" is Situation B. Say what that makes a passenger's warning worth. **PART B**
- Error analysis.** A group records these four drops in Situation A: 19.8 cm, 4.2 cm, 21.1 cm, 20.4 cm.
 - Calculate the reaction time the 4.2 cm drop by itself would give.
 - Name the **specific thing the catcher did** to produce a number that small. "Human error" is not an answer and earns zero.
 - Calculate the average reaction time with that drop included, and again with it thrown out and replaced by a fourth good drop of 20.1 cm. State which average is honest and why.
 - Compare the spread of this group's three good drops to the spread you recorded for yourself. Whose measurement is more trustworthy, and how do you know from the spread alone?

ERROR ANALYSIS

- The school zone.** You are driving through a school zone at **25 mph**. The posted limit is 20 — you are speeding. A child steps into the crosswalk ahead of you. You see them, you brake, and you stop. The question is *where* you stop.

QUANTITY	VALUE	WHERE IT COMES FROM
Your speed	25 mph	given — and constant until you touch the brake
Conversion	1 mile = 1609 m · 1 h = 3600 s	given
Braking distance, 25 mph → 0	8.9 m	dry asphalt, good tires, $a = -7.0 \text{ m/s}^2$
Width of the crosswalk	3.0 m	the child is standing in the middle of it

- Convert 25 mph to m/s. Carry the units through the conversion so you can see them cancel.
- Using your **Situation A** reaction time and $v = d \div t$ rearranged, find your **reaction distance** — how far the car travels before your foot even reaches the brake.
- Do part (b) again with your **Situation C** reaction time.
- Find the **total stopping distance** for each case: reaction distance + braking distance.
- Subtract. How many extra meters did the distracted version cost you? Compare that number to the 3.0 m crosswalk and write one sentence — not a paragraph — saying plainly what those extra meters mean for the child standing in the middle of it.

The push. Everything before your foot touches the brake happens at a constant 25 mph, because you have not braked yet. That is exactly why $v = d \div t$ works for the reaction distance and the kinematic equations do not — there is no acceleration during your reaction time. The braking distance is the same 8.9 m either way, because once the brakes are on the car does not care why it took you so long to get there. **The entire difference between your two totals is distance you covered while you were still deciding.**